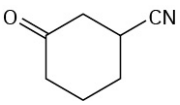
PROBLEM 19-20 The –OH group adds to the Re face at C2, and –H adds to the Re face at C3, to yield (2*R*,3*S*)-isocitrate.

PROBLEM 19-21



PROBLEM 19-22 **(a)** 3-Buten-2-one + $(CH_3CH_2CH_2)_2CuLi$ **(b)** 3-Methyl-2-cyclohexenone + $(CH_3)_2CuLi$ **(c)** 4-*tert*-Butyl-2-cyclohexenone + $(CH_3CH_2)_2CuLi$ **(d)** Unsaturated ketone + $(H_2C{=}CH)_2CuLi$

PROBLEM 19-23 Look for appearance of either an alcohol or a saturated ketone in the product.

PROBLEM 19-24 **(a)** 1715 cm^{-1} **(b)** 1685 cm^{-1} **(c)** 1750 cm^{-1} **(d)** 1705 cm^{-1} **(e)** 1715 cm^{-1} **(f)** 1705 cm^{-1}

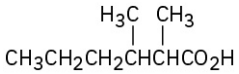
PROBLEM 19-25 **(a)** Different peaks due to McLafferty rearrangement **(b)** Different peaks due to α cleavage and McLafferty rearrangement **(c)** Different peaks due to McLafferty rearrangement

PROBLEM 19-26 IR: 1750 cm^{-1}; MS: 140, 84

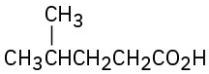
Chapter 20

PROBLEM 20-1 **(a)** 3-Methylbutanoic acid **(b)** 4-Bromopentanoic acid **(c)** 2-Ethylpentanoic acid **(d)** *cis*-4-Hexenoic acid **(e)** 2,4-Dimethylpentanenitrile **(f)** *cis*-1,3-Cyclopentanedicarboxylic acid

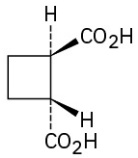
PROBLEM 20-2 **(a)**



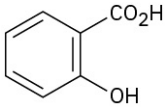
(b)



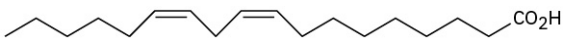
(c)



(d)



(e)



(f) $CH_3CH_2CH{=}CHCN$

PROBLEM 20-3 Dissolve the mixture in ether, extract with aqueous NaOH, separate and acidify the aqueous layer, and extract with ether.

PROBLEM 20-4 43%

PROBLEM 20-5 **(a)** 82% dissociation **(b)** 73% dissociation

PROBLEM 20-6 Lactic acid is stronger because of the inductive effect of the –OH group.

PROBLEM 20-7 The dianion is destabilized by repulsion between charges.

PROBLEM 20-8 More reactive

PROBLEM 20-9 **(a)** *p*-Methylbenzoic acid < Benzoic acid < *p*-Chlorobenzoic acid **(b)** Acetic acid < Benzoic acid < *p*-Nitrobenzoic acid

PROBLEM 20-10 **(a)** 1. Mg; 2. CO_2; 3. H_3O^+ **(b)** 1. Mg; 2. CO_2; 3. H_3O^+ or 1. NaCN; 2. H_3O^+

PROBLEM 20-11 1. NaCN; 2. H_3O^+; 3. $LiAlH_4$

PROBLEM 20-12 1. PBr_3; 2. NaCN; 3. H_3O^+; 4. $LiAlH_4$

PROBLEM 20-13 **(a)** Propanenitrile + CH_3CH_2MgBr, then H_3O^+ **(b)** *p*-Chlorobenzonitrile + CH_3MgBr, then H_3O^+

PROBLEM 20-14 1. NaCN; 2. CH_3CH_2MgBr, then H_3O^+

PROBLEM 20-15 A carboxylic acid has a very broad –OH absorption at 2500–3300 cm^{-1}.

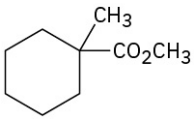
PROBLEM 20-16 4-Hydroxycyclohexanone: H–C–O absorption near 4 δ in the ^1H spectrum and C=O absorption near 210 δ in the ^{13}C spectrum. Cyclopentanecarboxylic acid: $-CO_2H$ absorption near 12 δ in the ^1H spectrum and $-CO_2H$ absorption near 170 δ in the ^{13}C spectrum.

Chapter 21

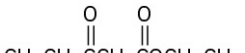
PROBLEM 21-1 **(a)** 4-Methylpentanoyl chloride **(b)** Cyclohexylacetamide **(c)** Isopropyl 2-methylpropanoate **(d)** Benzoic anhydride **(e)** Isopropyl cyclopentanecarboxylate **(f)** Cyclopentyl 2-methylpropanoate **(g)** *N*-Methyl-4-pentenamide **(h)** (*R*)-2-Hydroxypropanoyl phosphate **(i)** Ethyl 2,3-dimethyl-2-butenethioate

PROBLEM 21-2 **(a)** $C_6H_5CO_2C_6H_5$ **(b)** $CH_3CH_2CH_2CON(CH_3)CH_2CH_3$ **(c)** $(CH_3)_2CHCH_2CH(CH_3)COCl$

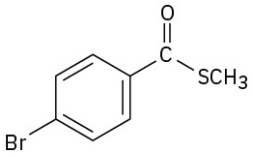
(d)



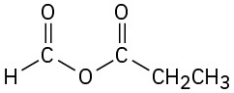
(e)



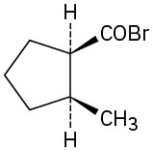
(f)



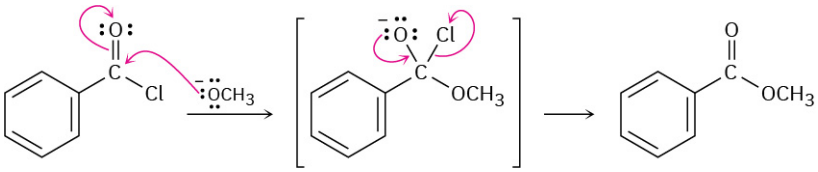
(g)



(h)



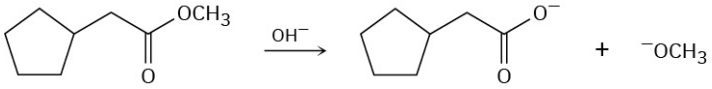
PROBLEM 21-3



PROBLEM 21-4 **(a)** Acetyl chloride > Methyl acetate > Acetamide **(b)** Hexafluoroisopropyl acetate > 2,2,2-Trichloroethyl acetate > Ethyl acetate

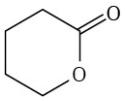
PROBLEM 21-5 **(a)** $CH_3CO_2^-\ Na^+$ **(b)** CH_3CONH_2 **(c)** $CH_3CO_2CH_3 + CH_3CO_2^-\ Na^+$ **(d)** $CH_3CONHCH_3$

PROBLEM 21-6



PROBLEM 21-7 **(a)** Acetic acid + 1-butanol **(b)** Butanoic acid + methanol **(c)** Cyclopentanecarboxylic acid + isopropyl alcohol

PROBLEM 21-8



PROBLEM 21-9 **(a)** Propanoyl chloride + methanol **(b)** Acetyl chloride + ethanol **(c)** Benzoyl chloride + ethanol

PROBLEM 21-10 Benzoyl chloride + cyclohexanol

PROBLEM 21-11 This is a typical nucleophilic acyl substitution reaction, with morpholine as the nucleophile and chloride as the leaving group.

PROBLEM 21-12 **(a)** Propanoyl chloride + methylamine **(b)** Benzoyl chloride + diethylamine **(c)** Propanoyl chloride + ammonia

PROBLEM 21-13 **(a)** Benzoyl chloride + $[(CH_3)_2CH]_2CuLi$, or 2-methylpropanoyl chloride + Ph_2CuLi **(b)** 2-Propenoyl chloride + $(CH_3CH_2CH_2)_2CuLi$, or butanoyl chloride + $(H_2C{=}CH)_2CuLi$

PROBLEM This is a typical nucleophilic acyl substitution reaction, with *p*-hydroxyaniline as the nucleophile